

# Optimization

## 3.1 Multivariate optimization: definition and general remarks

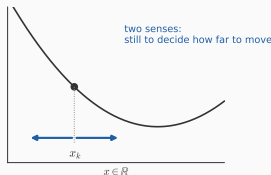
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José Firmino Aguilar Madeira

# From Chapter 2 to 3: what has changed?

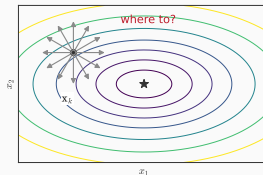
## Chapter 3. Unconstrained multivariate optimization

Ch. 2:  $x \in \mathbb{R}$



In 1D, the choice of direction reduces to **two options**,  $\leftarrow$  and  $\rightarrow$ ; once one is chosen (or the search interval), it remains to decide **how far to move**.

Ch. 3:  $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$

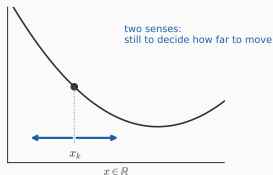


In  $n$ D, there are **infinitely many** possible directions from  $x_k$ .

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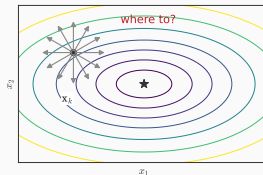
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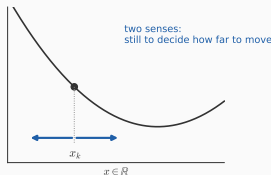


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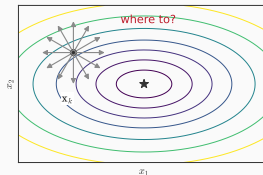
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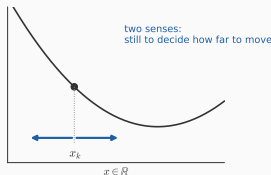
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In 1D, the choice of direction reduces to **two options**,  $\leftarrow$  and  $\rightarrow$ ; once one is chosen (or the search interval), it remains to decide **how far to move**.

**1st question:** where should we move?

**2nd question:** how far should we move in that direction?

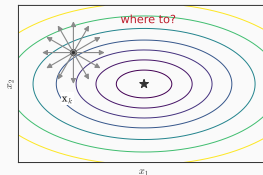
$d_k = \text{direction}$

$\alpha_k > 0 = \text{step}$

$\Rightarrow$

$x_{k+1} = x_k + \alpha_k d_k$

Ch. 3:  $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$



In  $n$ D, there are **infinitely many** possible directions from  $x_k$ .

# The multivariate problem: direction and step

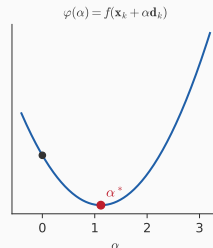
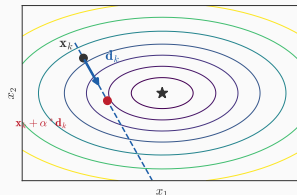
$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}), \quad \mathbf{x} = (x_1, \dots, x_n)^T,$$

unconstrained (constraints are left for Ch. 4).  
Maximizing  $f$  is minimizing  $-f$ .

**The methods of Ch. 2 do not disappear**

Once  $\mathbf{d}_k$  is chosen, the step solves a one-dimensional problem:

$\min_{\alpha > 0} \varphi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{d}_k)$  — the line search (golden section, Powell, Newton...).



Ch. 3 chooses the **direction**

Ch. 2 can choose the **step**

This is the form of direction-based methods (3.3; in part, Hooke–Jeeves); random search, Nelder–Mead and Box progress in a different way (3.2).

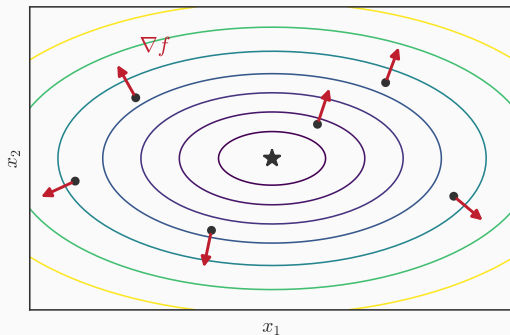
## Gradient: where does $f$ increase fastest?

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix} \quad (\text{R})$$

Along a unit direction  $\mathbf{d}$ ,  $f$  changes at the rate  $\nabla f(\mathbf{x})^T \mathbf{d} \leq \|\nabla f(\mathbf{x})\|$  (Cauchy–Schwarz), with equality for  $\mathbf{d} = \nabla f / \|\nabla f\|$ .

$\nabla f$  is normal to the level curve

$\nabla f$  points to the greatest local increase



$f(\mathbf{x}) = \frac{1}{2}x_1^2 + 2x_2^2$ : the gradient at six points.

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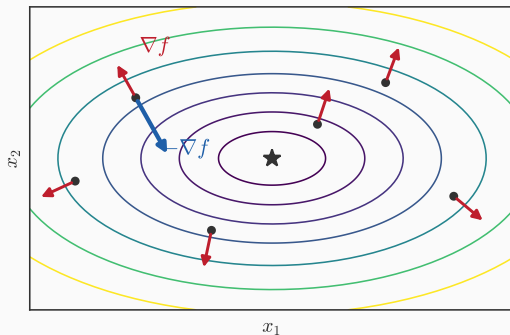
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To go down? The opposite direction:  $-\nabla f(\mathbf{x})$

(This is the idea of the steepest descent method, 3.3.1.)



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# What is a descent direction?

First-order Taylor ( $f$  differentiable), for  $\alpha > 0$ :

$$f(x + \alpha d) = f(x) + \alpha \nabla f(x)^T d + o(\alpha),$$

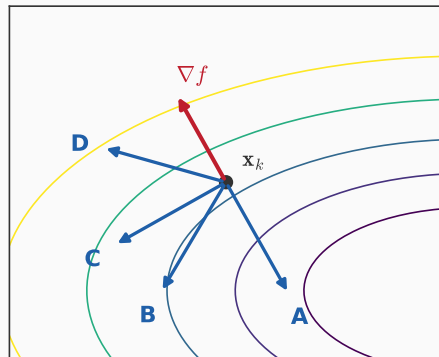
with  $o(\alpha)/\alpha \rightarrow 0$  as  $\alpha \rightarrow 0^+$ .

If  $\nabla f(x)^T d < 0$ , the linear term dominates for small  $\alpha$ :  $f(x + \alpha d) < f(x)$ .

## Descent direction

$d$  is a descent direction at  $x$  if  $\nabla f(x)^T d < 0$ :  
the angle between  $d$  and  $\nabla f$  is **greater than  $90^\circ$** .

**Question:** which of these arrows is a descent direction at  $x_k$ ?



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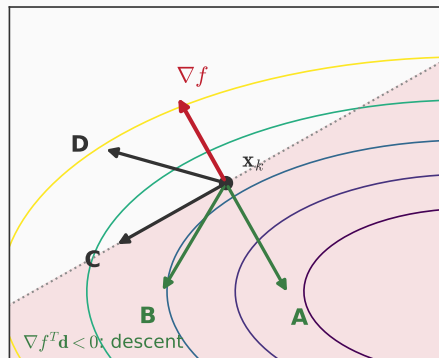
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**A** ( $-\nabla f$ ) and **B**: yes. **C**: tangent to the level curve,  $\nabla f^T d = 0$  — *not* a descent direction (first order does not decide). **D**:  $\nabla f^T d > 0$ , goes up.



## And when we can no longer go down?

Let  $x^*$  be a local minimum (unconstrained). If  $\nabla f(x^*) \neq 0$ , the direction  $d = -\nabla f(x^*)$  would give

$$\nabla f(x^*)^T d = -\|\nabla f(x^*)\|^2 < 0,$$

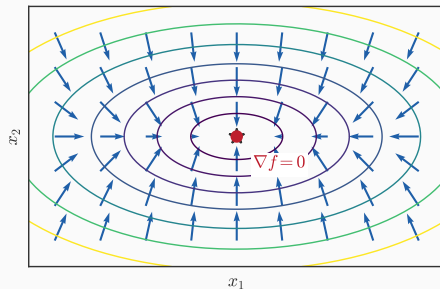
that is, a descent direction — and  $x^*$  would not be a minimum. Hence:

### First-order necessary condition

$$\begin{aligned} f \text{ differentiable at } x^* \text{ and } x^* \text{ local minimum} \\ \implies \nabla f(x^*) = 0. \end{aligned}$$

Points with  $\nabla f = 0$  are called **stationary points**.

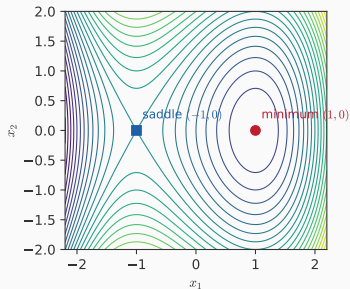
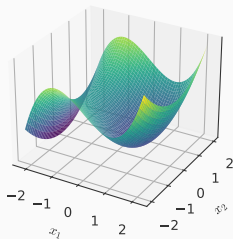
This is why  $\|\nabla f(x_k)\| \leq \varepsilon_g$  is a stopping criterion. With constraints (Ch. 4) this changes: at the minimum there may be descent directions that leave the feasible region.



Arrows:  $-\nabla f$ . Away from the minimum there is always a way down; at the minimum, there is not.

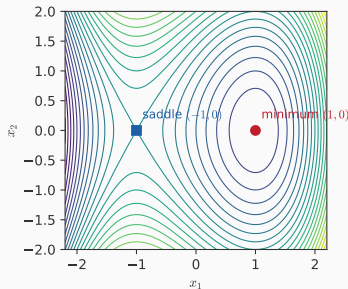
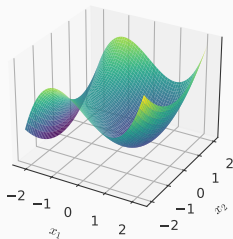
## But stationary $\neq$ minimum

$$f(x) = x_1^3 - 3x_1 + x_2^2: \quad \nabla f = \begin{pmatrix} 3x_1^2 - 3 \\ 2x_2 \end{pmatrix} = 0 \iff x \in \{(1,0), (-1,0)\}.$$



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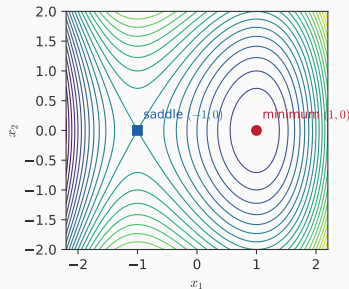
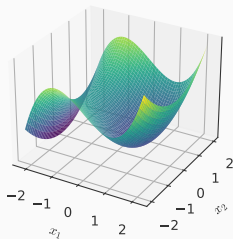
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Visually:  $(1,0)$  is a **minimum**;  $(-1,0)$  is a **saddle point** — it goes up along  $x_2$ , down along  $x_1$ .

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Visually:  $(1,0)$  is a **minimum**;  $(-1,0)$  is a **saddle point** — it goes up along  $x_2$ , down along  $x_1$ . **How do we tell them apart?** The gradient no longer helps (it is zero at both): we need the **curvature**  $\Rightarrow$  the Hessian.

## Hessian: the curvature

$$\nabla^2 f(x) = \left[ \frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{i,j=1}^n$$

symmetric if  $f \in C^2(\mathbb{R})$ .

Second-order Taylor:

$$\begin{aligned} f(x^* + s) &\approx f(x^*) + \nabla f(x^*)^T s \\ &\quad + \frac{1}{2} s^T \nabla^2 f(x^*) s. \end{aligned}$$

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At a stationary point the linear term vanishes:

$$f(x^* + s) - f(x^*) \approx \frac{1}{2} s^T \nabla^2 f(x^*) s$$

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| $\nabla^2 f(x^*)$                                           | curvature                            | $x^*$ is                                   |
|-------------------------------------------------------------|--------------------------------------|--------------------------------------------|
| pos. definite (eig. $> 0$ )                                 | upwards in all directions            | <b>strict local minimum</b>                |
| neg. definite (eig. $< 0$ )                                 | downwards in all directions          | strict local maximum                       |
| indefinite (mixed signs)                                    | up in one direction, down in another | saddle point                               |
| pos. semidefinite ( $\lambda_{\min} = 0 < \lambda_{\max}$ ) | zero in some direction               | not a maximum (inconclusive as to minimum) |
| all eigenvalues zero                                        | zero in all directions               | inconclusive                               |

**Second-order conditions** ( $f \in C^2$ ). Necessary: local minimum  $\Rightarrow \nabla^2 f(x^*)$  pos. semidefinite. Sufficient:  $\nabla f(x^*) = 0$  and  $\nabla^2 f(x^*)$  pos. definite  $\Rightarrow$  strict local minimum.

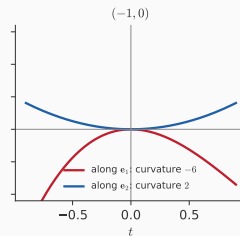
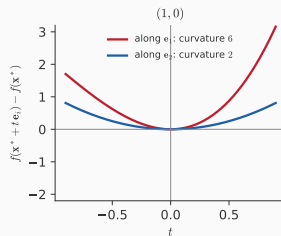
To check: eigenvalues or Sylvester (R).

# Complete example: minimum or saddle?

$$f(x) = x_1^3 - 3x_1 + x_2^2, \quad \nabla^2 f(x) = \begin{pmatrix} 6x_1 & 0 \\ 0 & 2 \end{pmatrix}.$$

- $(1, 0)$ :  $\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$ , eigenvalues  $6, 2 > 0$   
 $\Rightarrow$  **strict local minimum**,  $f = -2$ .
- $(-1, 0)$ :  $\begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}$ , eigenvalues  $-6, 2$   
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In the slices on the right, the curvature of each slice is the diagonal entry  $(e_i^T \nabla^2 f e_i)$ .

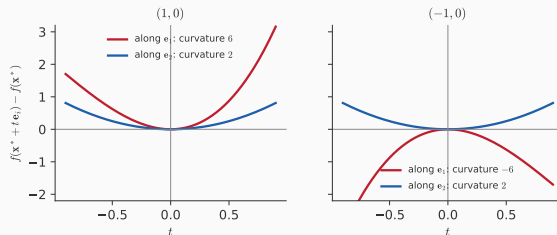


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**There is no global minimum:**  $f(x_1, 0) \rightarrow -\infty$  as  $x_1 \rightarrow -\infty$ .

Optimality conditions are **local**: they say what happens in a neighbourhood, not on the whole domain. Seeking the *global* minimum is another problem (Ch. 5).

# So how do we choose $d_k$ ? The map of methods

## Derivative-free (3.2) — only values $f(x)$

3.2.1 random search; 3.2.2 grid search;  
3.2.3 Nelder–Mead; 3.2.4 Box; 3.2.5 Hooke–Jeeves.

They make progress by comparing values of  $f$  (which does not automatically make them robust to noise, 3.2).

## Derivative-based (3.3) — $\nabla f$ , possibly $\nabla^2 f$

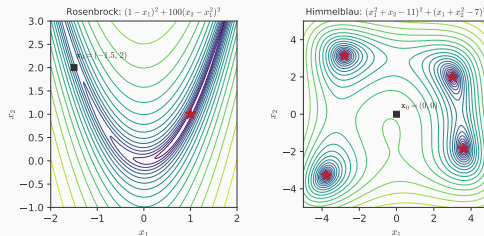
3.3.1 steepest descent:  $d = -\nabla f$   
3.3.2 Newton:  $d = -(\nabla^2 f)^{-1} \nabla f$   
3.3.3 conjugate gradients:  $-\nabla f$  corrected by the previous direction

**How much does the information cost?** By central differences (2.6), a gradient costs  $2n$  evaluations of  $f$  (forward:  $n$ , besides  $f(x)$ ) and a Hessian  $O(n^2)$ .

What information do we have about  $f$ ? How much does it cost to obtain? What can we guarantee?

For each method: what it uses, how it chooses the direction, how much it costs ( $n_f$ ,  $n_g$ ,  $n_H$ , counted separately, 2.1) and how it behaves in the Rosenbrock valley. At the end (3.3.3), all seven on the same problem, with the same starting point and the same criterion.

# Rosenbrock and Himmelblau: test tracks



**Rosenbrock**,  $f(x) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$ : a single minimum,  $(1, 1)$ , at the bottom of a **narrow, curved** valley — it punishes methods that move along the axes or along the gradient.

**Himmelblau**,  $f(x) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$ : four global minima, all with  $f = 0$  — the result depends on the starting point.

They are not examples to “solve”: they are **test tracks** to see how the algorithms behave.

## KEY TAKEAWAYS — 3.1

1. In direction-based methods, each iteration involves **two decisions**: the direction  $d_k$  and the step  $\alpha_k$ ; Ch. 2 can choose the step.
2.  $d$  is a **descent direction** if  $\nabla f(x)^T d < 0$ ; if  $\nabla f(x) \neq 0$ ,  $-\nabla f(x)$  is always one.
3. At a local minimum (unconstrained,  $f$  differentiable),  $\nabla f(x^*) = 0$  — but this only identifies **stationary points**.
4. At a stationary point, the **Hessian** classifies locally: positive definite  $\Rightarrow$  strict local minimum; indefinite  $\Rightarrow$  saddle.
5. The choice of method depends on the **available information**: without derivatives  $\rightarrow$  3.2; with derivatives  $\rightarrow$  3.3.

Next (3.2): **if we do not know  $\nabla f$ , how do we choose where to move?**

**References** Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 2.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996.

**Resources:** —